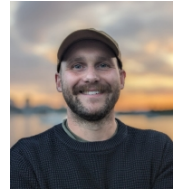


Jordan Sawchuk

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Education

- 2023 – ···· **Ph.D. Simon Fraser University** Nonequilibrium statistical physics
Thesis focus: *Differential geometry of dissipation in mesoscopic systems*
Supervisor: *Dr. David Sivak*
- 2020 – 2023 **B.Sc. University of British Columbia** Honours physics, minor in mathematics.
Thesis: *Quantization of constrained Hamiltonian systems and negative energy densities in QFT*.
Supervisor: *Dr. Daniel Vollick*
Graduating average: 97.1%
- 2018 – 2020 **A.S. Okanagan College** Science
Graduated with Distinction; Graduating average: 96.15%

Workshops

- 2025 **Statistical Aspects of Nonlinear Physics Summer School**, Institut des Hautes Études Scientifiques
- 2022 **PSI Start Program**, Perimeter Institute

Publications & Presentations

Journal Articles

- 1 **J. R. Sawchuk** and D. A. Sivak, “Global thermodynamic manifold for conservative control of stochastic systems,” *Physical Review Research*, vol. 8, no. 1, p. 013 004, 2026. [DOI: 10.1103/j59j-q88v](#).
- 2 **J. R. Sawchuk** and D. A. Sivak, “Thermodynamic geometry of friction on graphs: Resistance, commute times, and optimal transport,” *Physical Review E*, vol. 114, no. 2, p. L022105, 2026. [DOI: 10.1103/bhr1-lplt](#).

Invited Talks (Presenter marked with an asterisk)

- 1 **J. R. Sawchuk***, “A (nearly) random walk through thermodynamic geometry: Friction, optimal transport, and curvature,” SFU Mathematics of Computation, Application and Data Seminar, Burnaby, BC, 2026. [URL: https://researchseminars.org/seminar/AppliedMath](https://researchseminars.org/seminar/AppliedMath).

Contributed Posters & Talks

- 1 **J. R. Sawchuk*** and D. A. Sivak, “Equivalence of commute-time geometry, resistance distance, and thermodynamic geometry in slowly driven Markov chains,” Berkeley Statistical Mechanics Meeting, Berkeley, CA, 2026. [URL: https://berkeleystatmech.org/index.html](https://berkeleystatmech.org/index.html).
- 2 **J. R. Sawchuk*** and D. A. Sivak, “A global thermodynamic manifold for optimal conservative control,” APS Global Physics Summit, Anaheim, CA, 2025. [URL: https://summit.aps.org/events/MAR-W65/9](https://summit.aps.org/events/MAR-W65/9).
- 3 S. Alderson*, G. Cornell, A. Menard, I. Kennedy, E. Launer, A. Nikou, **J. R. Sawchuk**, A. Zavahir, J. Andrews, F. Moosvi, H. Nakahara, and J. Bobowski, “Jupyter-based physics labs: Introducing scientific computing & discovery,” UBC ALT-2040 Showcase, Kelowna, BC, 2024. [URL: https://github.com/UBC-Okanagan-Physics-Labs/ALT-2040-Showcase/](https://github.com/UBC-Okanagan-Physics-Labs/ALT-2040-Showcase/).

- 4 **J. R. Sawchuk***, I. Kennedy, E. Launer, A. Zavahir, A. Nikou, J. Andrews, H. Nakahara, F. Moosvi, and J. Bobowski, "Integrating discovery and computational skills into first-year physics labs," UBC ALT-2040 Showcase, Kelowna, BC, 2023.  URL: <https://github.com/UBC-Okanagan-Physics-Labs/Celebrate-Learning-Week-Poster/>.
- 5 **J. R. Sawchuk*** and J. Hopkinson, "Cation ordering on a lattice of corner-sharing tetrahedra: Investigation of a simple Ising-like model," UBC Faculty of Science Undergraduate Research Award Symposium, Kelowna, BC, 2022.

Awards & Funding

Fellowships & Scholarships

- 2025 – 2028  **NSERC CGS-D**, Simon Fraser University
- 2023  **NSERC CGS-M**, Simon Fraser University
-  **NSERC USRA**, Simon Fraser University
- 2022  **Irving K. Barber Undergraduate Research Award**, University of British Columbia
-  **James R. Conway Scholarship**, University of British Columbia
- 2021, 2022  **Deputy Vice-Chancellor Scholarship for Continuing Students**, University of British Columbia
- 2021  **Irving K. Barber Transfer Scholarship**, University of British Columbia

Academic Prizes

- 2023  **Graduating Student Award in Physics**, University of British Columbia
- 2022  **Upper-Level Physics Award**, University of British Columbia
- 2021  **Second-Year Physics Award**, University of British Columbia
- 2020  **President's Award**, Okanagan College
-  **Associate of Science Award**, Okanagan College

Travel Awards

- 2025, 2026  **Travel & Research Award**, Simon Fraser University

Research Experience

- 2026  **Research Contractor**, Impossible Fibers (*Speculative Technologies*)
Authored a technical report on protein-fiber computational devices, integrating a review of relevant literature with original modeling and analysis for transduction and reservoir computing.
- 2023  **Undergraduate Researcher**, Simon Fraser University (*NSERC USRA*)
Supervisor: *Dr. David Sivak*
- 2022  **Undergraduate Researcher**, University of British Columbia (*Irving K. Barber URA*)
Supervisor: *Dr. John Hopkinson*
- 2019 – 2020  **Microbiology Laboratory Assistant**, Okanagan College

Teaching & Mentoring

- 2022 – 2023  **Academic Assistant**, University of British Columbia
Rewrote the first-year physics laboratory manual; built Jupyter-based lab notebooks with guided exercises and data-analysis tutorials; designed student surveys for course evaluation.

Teaching & Mentoring (continued)

- 2021 – 2024  **Teaching Assistant**, University of British Columbia, Simon Fraser University
- 2020 – 2023  **Math & Physics Tutor**, Self-employed
- 2020 – 2021  **Writing Consultant**, University of British Columbia
- 2020  **Biology Laboratory Assistant**, Okanagan College
- 2019 – 2020  **Physics Learning Assistant**, Okanagan College
- 2015  **English Tutor**, Kamloops Immigration Centre

Service & Outreach

- 2024  **Conference Organizer**, Frontiers in Biophysics 2024
- 2022  **Volunteer Invigilator**, British Columbia Secondary School Mathematics Contest

Volunteering & Other Employment

Volunteering

- 2018 – 2022  **Farmhand**, Okanagan Fruit Tree Project Society (*Kelowna, BC*)
- 2017 – 2018  **Dog Walker**, BC SPCA (*Prince George, BC*)

Other Employment

- 2019 – 2021  **Farmhand**, Lightfoot Farm (*Kelowna, BC*)
- 2018 – 2019  **Helicopter & Drill Pad Construction**, BCR (*Whitehorse, YT*)
- 2017 – 2018  **Bartender**, Alpine Pub & Grill (*Prince George, BC*)
- 2016 – 2017  **Line Cook**, Munin's Post (*Kelowna, BC*)
- 2016  **Dispatcher**, BC Wildfire (*Kamloops, BC*)
- 2015  **Regional Lead Landscaper**, Class Design (*Kamloops, BC*)
- 2014 – 2015  **Breakfast Cook**, Fireside Grill (*Kamloops, BC*)
- 2013  **Sous Chef**, Long John's Pub (*Vernon, BC*)